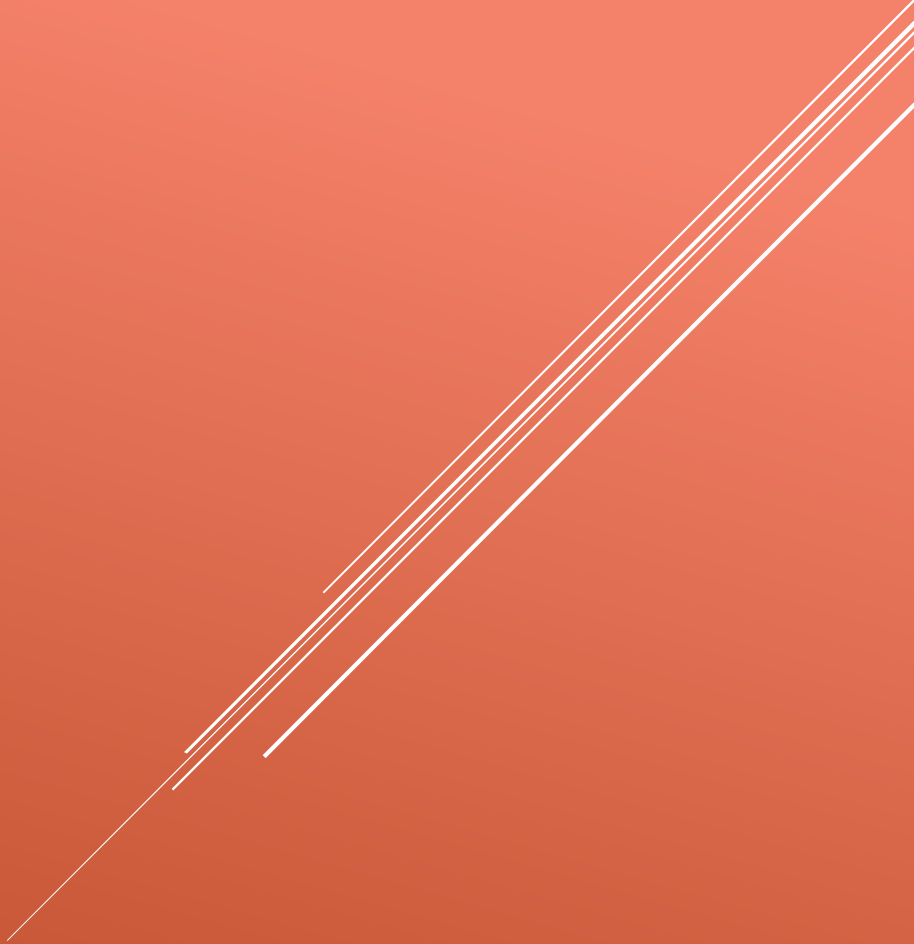




NOVATED LEASE PORTFOLIO ANALYSIS

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Executive Summary

Motor Insurer Inc. has recently acquired a portfolio of motor insurance policies for vehicles under novated lease agreements. This report reviews the portfolio's performance to date, evaluates its claims and premium experience by segment, and provides actionable recommendations to improve risk management and pricing adequacy.

Key findings:

- *Total Premiums Earned: \$619,142*
- *Total Net Claims Incurred to Date: \$468,168*
- *Ultimate Net Incurred Claims (after development): \$576,014*
- *Current Portfolio Loss Ratio: 0.76*
- *Ultimate Portfolio Loss Ratio: 0.93*
- *Total Policy Rows: 10,529*
- *Unique Vehicles: 1,969*
- *Claims Reported: 143*

Segmentation by age band and vehicle sum insured revealed significant variation in loss performance. While many segments remain profitable, some (e.g. 3C, 3E, 4D) exhibit significantly higher than average loss ratios and require corrective pricing or underwriting action.

The use of development factors to estimate ultimate incurred claims revealed that the portfolio is likely under-reserved in certain segments. The projected deterioration of loss ratios in several areas reinforces the need for risk-based pricing and continuous monitoring.

Recommendations include re-pricing high-loss segments, incentivizing retention in profitable groups, enhancing data capture (e.g. on finalization lags), and refining premium models beyond just age and sum insured.

1. Background and Objectives

Motor Insurer Inc. distributes insurance for vehicles acquired under novated leases via a partnering finance company. These policies are valid for up to one year unless cancelled earlier, with renewals available post-expiry. Premium and policy details are shared monthly via bordereaux files. Currently, premiums are priced based solely on driver age band (18 – 24, 25 – 30, 31 – 40, 41 – 50, 51+) & vehicle sum insured band (< \$40k, \$40k – < \$60k, \$60k – < \$80k, \$80k – < \$100k, \$100k+).

The key aims of this analysis are:

- To evaluate the *current portfolio experience*
- To identify high-risk or underperforming *segments*
- To assess loss ratios and *ultimate claims exposure*
- To provide *strategic recommendations* for improving profitability

2. Methodology & Segmentation Approach

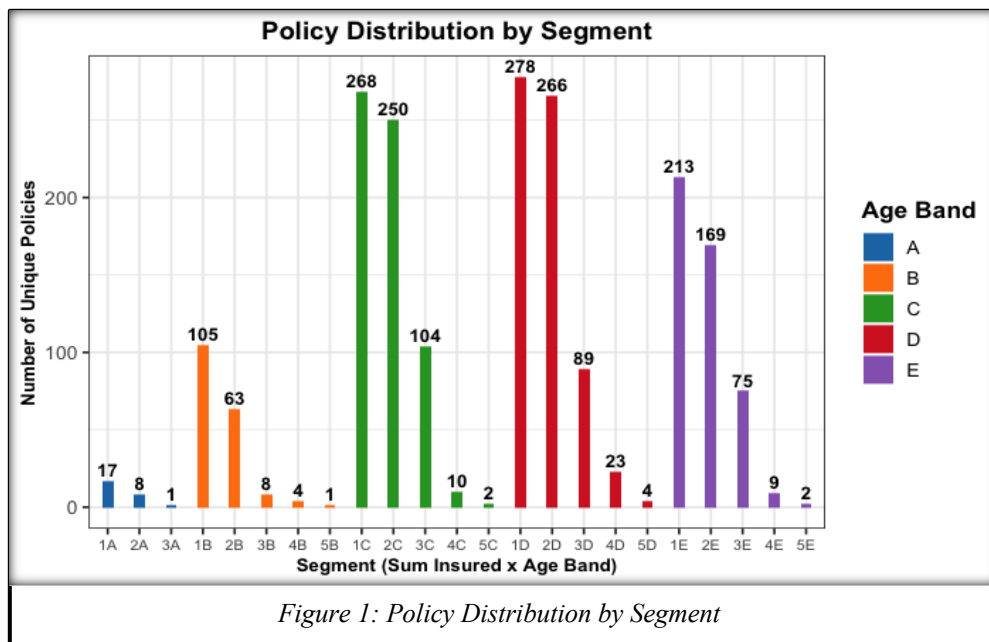
To assess the health and performance of the portfolio, I adopted a segmentation-based approach reflecting the current pricing model, which uses only two rating variables: driver age and vehicle sum insured. These were grouped into five *age bands (A–E)* and five *sum insured categories (1–5)*, creating 25 potential combinations (segments), labeled as 1A to 5E. Each segment represents a unique combination of risk profile based on age and asset value, allowing for more granular analysis of claims experience and premium adequacy.

The steps involved were:

- *Mapped* all policies into respective Age and Sum Insured bands.
- *Created a segment code* by concatenating Sum Insured Category (1–5) and Age Band (A–E).
- *Aggregated exposure, claims, and premium* values per segment.
- *Calculated key metrics*: Reported loss ratio, developed ultimate claims using development factors, and ultimate loss ratios.
- *Benchmarked* performance of each segment against a portfolio loss ratio target of 75%.

From a total of **1,969 unique policies**, segment distribution was highly uneven. The chart below shows that:

- The largest concentrations were in segments *1D* (278 policies) and *1C* (268 policies).
- Segments in age band *D* (31–40) and sum insured bands 1 – 3 (<\$80k) together accounted for over 60% of the total portfolio.
- Very few policies were written in segments like *5E* (2 policies) and *3A* (1 policy) — indicating sparse exposure in high sum insured, younger driver categories.

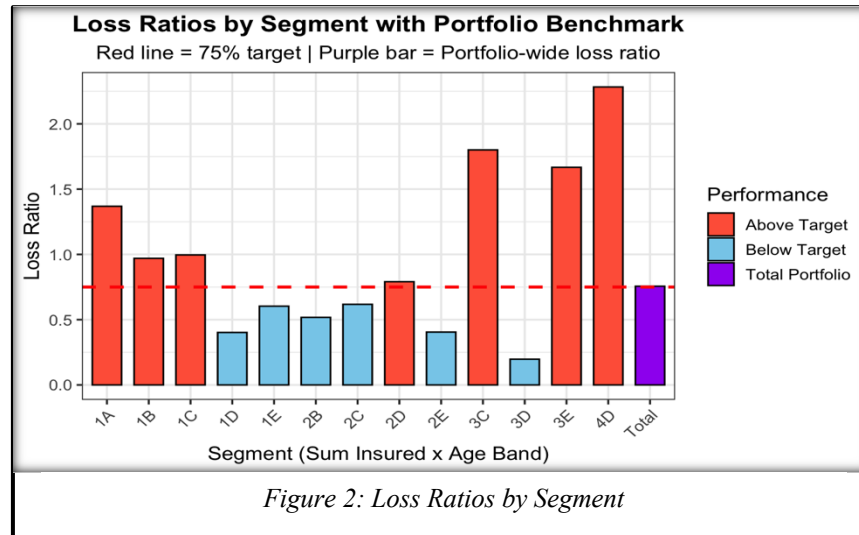


This segmentation formed the backbone of the entire analysis, enabling the identification of underperforming groups, pricing inefficiencies, and pockets of strong profitability that may warrant further focus or incentives.

3. Key Result & Analysis

3.1 Segment Loss ratio Performance

Segments with a high number of policies and loss ratios exceeding the 75% benchmark were flagged for pricing review.

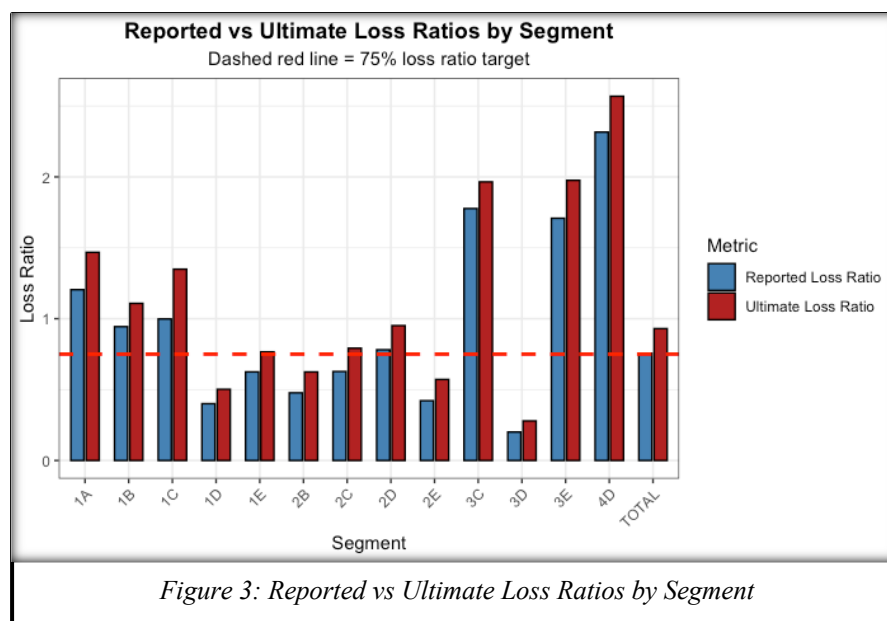


Key Observations:

- Over half the segments (7 of 13) breached the portfolio's 75% loss ratio target, with 3C, 3E, and 4D notably exceeding a 1.5 ratio, indicating potential underpricing risks.
- Several segments (1D, 1E, 2B, 2C, 2E, 3D) showed profitable outcomes, with no claims reported in some other segments, reflecting either low exposure to risk or favorable claim behavior.
- Despite high exposure, segment 2D hovered just above the threshold, while the overall portfolio recorded a loss ratio of 75.62%, closely aligned with the performance target.

3.2 Reported vs Ultimate Loss Ratios

To better estimate the true cost of claims, I applied monthly development factors to project ultimate incurred claims for each segment. This allowed a comparison between reported loss ratios (claims incurred to date) and their likely final values.



Key Observations:

- Almost every segment with claims saw their loss ratio deteriorate after applying development factors, highlighting the portfolio's sensitivity to claims finalization lags.
- For instance, *1A* rose from 1.20 to 1.47, *1C* from 1.00 to 1.35, and *2D* from 0.78 to 0.95, all exceeding or approaching the 0.75 benchmark post-adjustment.
- Segments with already poor experience worsened significantly: *3C* jumped from 1.78 to 1.96, *3E* from 1.71 to 1.98, and *4D* from 2.32 to 2.57, reinforcing their classification as loss-heavy outliers.
- The overall shift from reported (0.76) to ultimate loss ratio (0.93) reflects a 22% deterioration, indicating that delayed claims development is a non-trivial risk & need to reassess IBNR Reserves.

3.3 Segment Level Pricing Adequacy

The chart below visualizes the average monthly premium by segment, color-coded by ultimate loss ratio. While one might expect higher-risk segments to be charged more, the current pricing structure reveals notable mismatches between premium levels and actual claims experience.

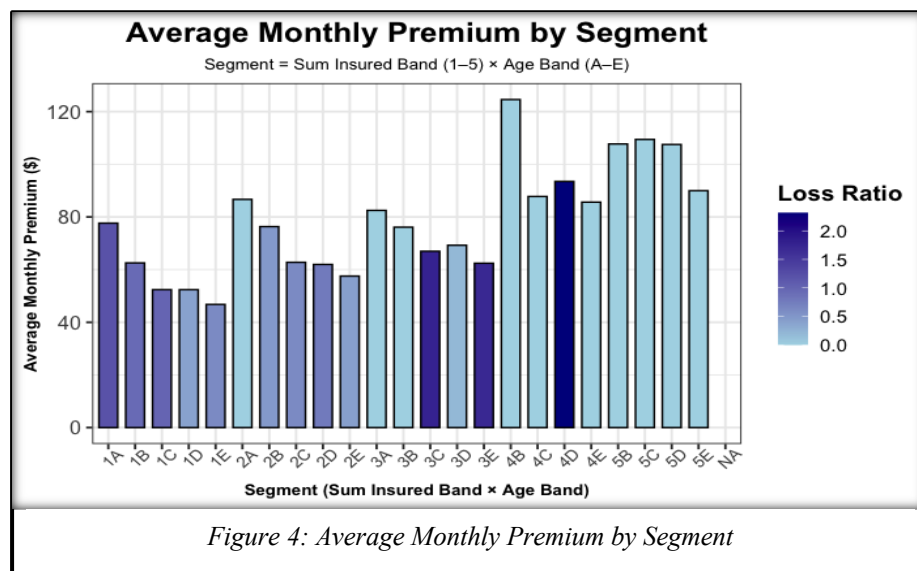


Figure 4: Average Monthly Premium by Segment

Key Observations:

- Segment *4B* has the highest premium at \$124.59, yet recorded no claims, indicating possible overpricing.
- Segment *3C*, despite a high ultimate loss ratio of 1.96, is charged only \$69.26, suggesting underpricing of high-risk profiles.
- Several costly segments (e.g. *5B–5D*) show no claims, raising concerns about pricing fairness for low-risk drivers.
- A general trend is observed: younger drivers are charged more within the same sum insured category, and premiums rise progressively with higher sum insured bands — consistent with intuitive pricing, though not always aligned with actual outcomes.

Overall, the **portfolio displays uneven performance, with high exposure concentrated in mid-age and mid-value vehicle segments**. Several segments, notably *3C*, *3E*, and *4D*, show ultimate loss ratios far exceeding the 75% target, highlighting pricing inadequacy. The 22% deterioration from reported to ultimate loss ratio signals **reserving risk**. Additionally, **some segments with high loss ratios are underpriced, while others with zero claims lack sufficient exposure to assess profitability**.

4. Strategic Recommendations

Based on the segment-level analysis, the following strategic actions are proposed to enhance portfolio performance and align with the target loss ratio of 75%:

- **Re-price unprofitable segments:** Segments like 3C (age 31–40, \$60k–\$80k vehicles), 3E (same age group, \$80k–\$100k), and 4D (age 51+, \$80k–\$100k) show ultimate loss ratios of 1.96, 1.98, and 2.57 respectively. These require urgent premium adjustments or tighter underwriting.
- **Optimize large-volume segments:** 2D (age 41–50, \$60k–\$80k), the most populated segment, has an ultimate loss ratio of 0.95. Even minor pricing tweaks here could yield major financial impact.
- **Retain low-risk groups:** Segments like 1D (age 18–24, <\$40k vehicles) and 3D (age 31–40, \$80k–\$100k) show strong profitability (ultimate ratios of 0.50 and 0.28). Retention incentives like renewal bonuses or no-claims discounts should be explored.

To support long-term portfolio growth and profitability, the following innovations could also be considered:

- **Implement Real-Time Pricing Flags:** Establish dynamic monitoring dashboards that flag segments exceeding a set loss ratio threshold (e.g. >1.0) within the past quarter. For instance, segment 2D currently sits at an ultimate loss ratio of 0.95, early intervention via quarterly tracking could prevent it from crossing the profitability threshold as volumes grow.
- **Explore Behavioral Pricing Incentives:** Introduce a telematics-based add-on (e.g. mobile app integration) to collect driving data and offer discounts for safe behavior (e.g. no hard braking, consistent speed). Target groups like 3C and 3E, with high loss ratios (1.96 and 1.98) could benefit from behavioral self-selection, reducing claims frequency over time while appealing to risk-conscious customers.

Conclusion

This report offers a data-driven diagnostic of Motor Insurer Inc.'s novated lease portfolio, revealing both promising segments and key underwriting vulnerabilities. Through segment-level analysis aligned with existing pricing variables, critical areas like 3C, 3E, and 4D were identified for immediate pricing correction, while profitable groups such as 1D and 3D offer opportunities for growth and retention. With a total ultimate portfolio loss ratio of 93%, now marginally above the target 75%, focused recalibration of high-volume and high-loss segments could yield substantial gains. Looking ahead, embedding real-time monitoring and behavioral-based pricing will not only improve responsiveness but also position the portfolio for long-term, sustainable profitability in a competitive market. This balanced strategy demonstrates both immediate action and future vision — essential traits for a portfolio ready to scale.

APPENDIX

Loss Ratio Table

<u>SEGMENT</u>	<u>LOSS RATIO</u>	<u>ULTIMATE LOSS RATIO</u>
1A	1.204	1.468
1B	0.944	1.108
1C	0.998	1.349
1D	0.401	0.502
1E	0.625	0.766
2A	0.000	0.000
2B	0.477	0.624
2C	0.628	0.792
2D	0.781	0.951
2E	0.422	0.572
3A	0.000	0.000
3B	0.000	0.000
3C	1.777	1.965
3D	0.200	0.279
3E	1.708	1.976
4B	0.000	0.000
4C	0.000	0.000
4D	2.316	2.569
4E	0.000	0.000
5B	0.000	0.000
5C	0.000	0.000
5D	0.000	0.000
5E	0.000	0.000
TOTAL	0.756	0.930

Additional EDA

